

Probability and Stochastics for Finance

A Report based on NPTEL Course by Dr. Joydeep Dutta (IIT Kanpur)

Harsh Modak

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Overview of Week 1

The first week of the course lays the essential mathematical and conceptual groundwork for quantitative finance. The lectures begin with the core principle of the time value of money, exploring various methods of interest rate calculation. This foundation is then applied to the valuation of fixed-income securities, primarily bonds. The focus then shifts to the term structure of interest rates, or the yield curve, a critical indicator in financial markets. Finally, the week introduces the powerful framework of mathematical optimization for financial decision-making and provides a crash course on the Karush-Kuhn-Tucker (KKT) conditions, the primary tool for solving such constrained optimization problems.

1 Fundamentals of Interest Rates

This topic covers the core concept of the time value of money, quantifying how the value of an investment changes over time.

- **Simple Interest:** Future value (A) is calculated on the principal (P) at rate (r) over time (t).

$$A = P(1 + rt)$$

- **Compound Interest:** Interest is calculated on the principal plus accumulated interest, compounded n times per year.

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

- **Continuous Compounding:** The theoretical limit of compounding as the frequency approaches infinity, widely used in financial modeling.

$$A = Pe^{rt}$$

- **Present Value (PV):** The current value of a future sum of money, found by discounting it back to the present.

$$PV = A(1 + r)^{-t} \quad \text{or} \quad PV = Ae^{-rt}$$

- **Effective Annual Rate (EAR):** The actual annual rate of interest earned after accounting for compounding.

$$EAR = \left(1 + \frac{r}{n} \right)^n - 1$$

2 Fixed Income Securities

This topic focuses on the valuation of debt instruments. The price of a bond is the sum of the present values of all its expected future cash flows.

- **Bond Price Formula:** The price (P) is the sum of discounted periodic coupons (C) and the face value (F) at maturity (T), using the yield to maturity (y) as the discount rate.

$$P = \sum_{t=1}^T \frac{C}{(1+y)^t} + \frac{F}{(1+y)^T}$$

- **Zero-Coupon Bond:** A bond that pays no coupons. Its price is simply the discounted face value.

$$P = \frac{F}{(1+y)^T} \quad \text{or} \quad P = Fe^{-yT}$$

3 Term Structure of Interest Rates – I

This topic describes the relationship between interest rates and their time to maturity (the yield curve), using spot and forward rates.

- **Spot Rate (s_t):** The annualized yield on a zero-coupon bond maturing at time t .
- **Forward Rate (f_{t_1, t_2}):** The interest rate for a future period between t_1 and t_2 , locked in today.
- **No-Arbitrage Relation:** This fundamental principle ensures that risk-free profits cannot be made by trading between bonds of different maturities. It links spot and forward rates.

$$(1 + s_2)^2 = (1 + s_1)(1 + f_{1,2})$$

- **Bootstrapping:** An iterative method to derive the zero-coupon spot rate curve from the market prices of coupon-paying bonds.

4 Optimization Models in Finance

This topic introduces mathematical optimization to find the best allocation of resources. Markowitz's Mean-Variance model seeks the optimal trade-off between risk and return.

- **Objective Function:** Minimize portfolio risk, measured by variance (σ_p^2). $\mathbf{w}$ is the vector of asset weights and Σ is the covariance matrix of asset returns.

$$\min_{\mathbf{w}} \quad \sigma_p^2 = \mathbf{w}^T \Sigma \mathbf{w}$$

- **Constraints:** The portfolio must meet a target expected return (R^*) and the weights must sum to one. μ is the vector of expected returns.

$$\mathbf{w}^T \mu = R^*$$

$$\mathbf{w}^T \mathbf{1} = 1$$

5 Crash Course on KKT Conditions

This topic presents the Karush-Kuhn-Tucker (KKT) conditions, a powerful set of necessary conditions for identifying optimal solutions in constrained nonlinear optimization problems.

- **Problem Formulation:** A general optimization problem with inequality (g_i) and equality (h_j) constraints.

$$\min_x f(x) \quad \text{s.t.} \quad g_i(x) \leq 0, \quad h_j(x) = 0$$

- **Lagrangian Function:** This function combines the objective function and the constraints into a single equation using Lagrange multipliers (λ_i, μ_j).

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \mu_j h_j(x)$$

- **KKT Conditions:** A set of four conditions that must hold at the optimal solution.
 1. **Stationarity:** $\nabla_x \mathcal{L}(x, \lambda, \mu) = 0$
 2. **Primal Feasibility:** $g_i(x) \leq 0, \quad h_j(x) = 0$
 3. **Dual Feasibility:** $\lambda_i \geq 0$
 4. **Complementary Slackness:** $\lambda_i g_i(x) = 0$ for all i .

Overview of Week 2

Week 2 delves deeply into the Mean-Variance Portfolio Optimization framework. The lectures build from the basic two-asset case to the multi-asset scenario, introducing the concept of the efficient frontier. The analysis is then extended by incorporating a risk-free asset, leading to the Capital Market Line (CML). Finally, the week culminates in the development of the Capital Asset Pricing Model (CAPM), a cornerstone of modern finance for pricing securities.

6 Mean Variance Portfolio Optimization I & II

These lectures introduce the foundational principles of modern portfolio theory. A **portfolio** is a collection of financial assets, such as stocks and bonds. The goal is to combine these assets to achieve a desired balance between risk and reward.

- **Reward and Risk:** In this framework, "reward" is measured by the portfolio's **expected return** ($E[R_p]$), while "risk" is measured by its **variance** (σ_p^2) or standard deviation (σ_p).

$$E[R_p] = \sum_{i=1}^N w_i E[R_i] = \mathbf{w}^T \boldsymbol{\mu}$$

- **Diversification and Portfolio Risk:** The key insight is that portfolio risk is not just the average of individual asset risks. Through **diversification**, combining assets that do not move perfectly together can reduce overall portfolio variance. This interaction is captured by covariance.
- **Variance-Covariance Matrix (Σ):** This matrix is essential for calculating portfolio risk. Its diagonal elements are the variances (σ_i^2) of each asset, and its off-diagonal elements are the covariances ($\text{Cov}(R_i, R_j)$) between each pair of assets.
- **Portfolio Variance Formula:** For a two-asset portfolio, the variance is:

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \text{Cov}(R_1, R_2)$$

For an N-asset portfolio, the general matrix form is:

$$\sigma_p^2 = \mathbf{w}^T \Sigma \mathbf{w}$$

- **The Mean-Variance Model:** The model seeks to find the set of optimal portfolios by solving a **Quadratic Programming (QP)** problem with affine (linear)

constraints. When **short selling** (selling a borrowed asset) is allowed, weights (w_i) can be negative. If short selling is disallowed, a non-negativity constraint is added:

$$w_i \geq 0 \quad \text{for all } i$$

- **Efficient Frontier:** The solution to the model is the efficient frontier—a curve representing the set of portfolios that offer the highest expected return for each level of risk.
- **Two-Fund Theorem:** This theorem states that, when short selling is allowed, any portfolio on the efficient frontier can be replicated by a combination of any two other portfolios on the frontier.

7 Mean Variance Portfolio Optimization III

This topic introduces a risk-free asset into the portfolio construction problem, dramatically changing the set of investment opportunities.

- **Risk-Free Asset:** An asset with a certain return r_f and zero variance ($\sigma_f = 0$).
- **Capital Market Line (CML):** When a risk-free asset is available and short selling of risky assets is permitted, the new efficient frontier becomes a straight line. This line starts at the risk-free rate and is tangent to the original risky efficient frontier. All efficient portfolios are now combinations of the risk-free asset and a single "tangency portfolio" of risky assets.
- **Equation of the CML:** It describes the expected return of an efficient portfolio ($E[R_p]$) based on its risk (σ_p).

$$E[R_p] = r_f + \left(\frac{E[R_T] - r_f}{\sigma_T} \right) \sigma_p$$

- **Sharpe Ratio:** The slope of the CML, $\frac{E[R_T] - r_f}{\sigma_T}$, which measures the excess return per unit of risk. The tangency portfolio (T) is the risky portfolio that maximizes this ratio.

8 Mean Variance Portfolio Optimization IV

This lecture develops the Capital Asset Pricing Model (CAPM), an equilibrium model that describes the expected return on any asset.

- **Capital Asset Pricing Model (CAPM):** In equilibrium, all investors hold the same tangency portfolio, which becomes the "market portfolio" (M). The CAPM states that the expected return on any security ($E[R_i]$) depends on the risk-free rate, the expected market return, and the security's systematic risk, measured by Beta (β_i). This derivation assumes investors can take long or short positions.

$$E[R_i] = r_f + \beta_i(E[R_M] - r_f)$$

- **Beta (β_i):** A measure of a security's volatility relative to the overall market. It quantifies the systematic risk that cannot be eliminated by diversification.

$$\beta_i = \frac{\text{Cov}(R_i, R_M)}{\text{Var}(R_M)}$$

9 Last lecture on Portfolio Optimization

This final lecture summarizes the portfolio optimization framework and discusses its critical assumptions and practical limitations.

- **Key Assumptions:** The model assumes investors are rational and risk-averse, returns are normally distributed, there are no taxes or transaction costs, and all investors have access to the same information and can borrow, lend, and short sell without restriction.
- **Limitations:** In practice, these assumptions are often violated. Returns may not be normal, correlations can be unstable, and estimating expected returns accurately is notoriously difficult. This highlights the gap between the theoretical model and real-world application.

Overview of Week 3

Week 3 transitions from portfolio theory to the fundamentals of derivative pricing. It begins by formalizing the CAPM through the lens of the One-Fund Theorem. The core of the week is dedicated to introducing the Binomial Model (Cox-Ross-Rubinstein model), a powerful tool for pricing options. The lectures build this model from a single period to multiple periods, introducing critical concepts like no-arbitrage, hedging, complete markets, risk-neutral probability, and the connection between asset pricing and martingale processes.

10 Capital Asset Pricing Model and the One-Fund Theorem

This topic formalizes a key insight from the introduction of a risk-free asset into portfolio theory.

- **The One-Fund Theorem:** A direct consequence of the CML, this theorem states that all rational investors will hold a combination of the risk-free asset and a single optimal portfolio of risky assets (the tangency portfolio).
- **Implication:** The investment decision is separated into two parts: (1) identifying the single optimal risky fund (the market portfolio in CAPM equilibrium), and (2) allocating capital between this fund and the risk-free asset based on individual risk preference.

11 The Binomial Model I: Single-Period Option Pricing

This introduces the binomial model for pricing a derivative, whose value depends on an underlying asset like a stock.

- **Model Setup:** A stock price S_0 can move to $S_u = S_0u$ (up-state) or $S_d = S_0d$ (down-state) in one period. The no-arbitrage condition requires $d < 1 + r_f < u$.
- **European Call Option Payoff:** The value of a call option with strike K at maturity T is:

$$C_T = \max(S_T - K, 0)$$

This results in payoffs C_u in the up-state and C_d in the down-state.

- **Replicating Portfolio:** The core idea is to create a portfolio of Δ shares of the stock and a cash amount B in a risk-free bond that exactly matches the option's payoffs.

$$\Delta S_u + B(1 + r_f) = C_u$$

$$\Delta S_d + B(1 + r_f) = C_d$$

- **Hedge Ratio (Δ):** Solving the system gives the number of shares needed to hedge the option position.

$$\Delta = \frac{C_u - C_d}{S_u - S_d}$$

- **Bond Position (B):** This is the amount of cash to borrow or lend.

$$B = \frac{C_d - \Delta S_d}{1 + r_f} = \frac{uC_d - dC_u}{(u - d)(1 + r_f)}$$

- **No-Arbitrage Price:** The option's price today, C_0 , must equal the cost of the replicating portfolio to prevent risk-free profits.

$$C_0 = \Delta S_0 + B$$

12 The Binomial Model II: Risk-Neutral Valuation

This section presents a more direct pricing method by deriving a synthetic probability measure.

- **Complete Market:** The single-period binomial model is a complete market because any derivative payoff can be perfectly replicated.
- **Risk-Neutral Probability (q):** By substituting the formulas for Δ and B into the no-arbitrage price equation $C_0 = \Delta S_0 + B$, we can rearrange the terms to define a special probability q . This is the probability that would make the expected return on the stock exactly equal to the risk-free rate r_f .

$$q = \frac{(1 + r_f) - d}{u - d} \quad \text{and} \quad (1 - q) = \frac{u - (1 + r_f)}{u - d}$$

- **Risk-Neutral Pricing Formula:** The option price is the expected value of its future payoffs, calculated with the risk-neutral probability q and discounted at the risk-free rate. This elegant formula is algebraically equivalent to the replication

cost.

$$C_0 = \frac{1}{1 + r_f} [qC_u + (1 - q)C_d]$$

13 The Binomial Model III: The Multi-Period Model

The single-period model is extended to multiple steps, creating a binomial tree for the stock price path.

- **Option Price Computation:** The price is found by working backward from the final payoffs. At each node in the tree, the option's value is the discounted expected value of its two possible future values, using the same risk-neutral probability q .
- **General Formula:** For an option with N periods, its price is the discounted sum of all possible terminal payoffs, weighted by their risk-neutral probabilities.

$$C_0 = \frac{1}{(1 + r_f)^N} \sum_{j=0}^N \left[\binom{N}{j} q^j (1 - q)^{N-j} \right] \max(S_0 u^j d^{N-j} - K, 0)$$

Here, $\binom{N}{j} q^j (1 - q)^{N-j}$ is the risk-neutral probability of reaching the terminal node with j up-moves and $N - j$ down-moves.

14 The Binomial Model IV: The Martingale Approach to Pricing

This connects pricing theory to the mathematical concept of a martingale, providing a deep theoretical foundation.

- **Portfolio Wealth Process (V_t):** This describes the value of a portfolio over time. A trading strategy consists of holding Δ_t units of the stock and B_t units of the risk-free asset (bond) at time t . The portfolio's value is:

$$V_t = \Delta_t S_t + B_t$$

- **Self-Financing Strategy:** A trading strategy is self-financing if changes in the portfolio's value are due only to gains or losses on the assets, with no external funds added or withdrawn. This means the value of the portfolio before rebalancing at time $t + 1$ is equal to its value after rebalancing at time t .

$$\Delta_{t+1} S_{t+1} + B_{t+1} = \Delta_t S_{t+1} + B_t (1 + r_f)$$

- **Martingale Process:** A stochastic process X_t where the best forecast of a future value, given all information up to time t , is its current value.

$$E_t[X_{t+1}] = X_t$$

- **The Martingale Pricing Principle:** In a complete, no-arbitrage market, there exists a unique risk-neutral probability measure Q . Under this measure, the **discounted** wealth process of any self-financing portfolio (including individual assets) is a martingale.

$$\frac{V_t}{(1 + r_f)^t} = E_t^Q \left[\frac{V_{t+1}}{(1 + r_f)^{t+1}} \right]$$

This is the cornerstone of modern asset pricing. It implies that the price of a derivative is the discounted expected value of its future payoff under the risk-neutral measure, because the derivative's price process must also be a martingale.

Overview of Week 4

The final week bridges the gap between discrete-time models and the continuous-time framework that is the standard for modern financial engineering. It introduces Girsanov's Theorem, the fundamental tool for changing probability measures, which is essential for risk-neutral pricing in continuous time. This leads to modeling stock prices with Geometric Brownian Motion and culminates in the derivation of the celebrated Black-Scholes formula for option pricing. The course concludes by stating the Martingale Representation Theorem, which underpins the concept of hedging in continuous-time markets.

15 Girsanov's Theorem: The Basic Tool

This lecture introduces the mathematical tools required to change from the real-world probability measure (P) to the risk-neutral measure (Q).

- **Radon-Nikodym Derivative:** This concept formalizes the change of measure. If a measure Q is absolutely continuous with respect to P , there exists a random variable $\eta = \frac{dQ}{dP}$ such that for any event A , the probability under Q can be found by taking the expectation under P :

$$Q(A) = E_P[1_A \eta]$$

where 1_A is the indicator function for event A .

- **Key Lemmas for Proof:** The lecture introduces important preliminary results, particularly concerning exponential martingales, which are crucial for constructing the Radon-Nikodym derivative process used in Girsanov's theorem.

16 Girsanov's Theorem: Statement and Proof

This section provides the formal statement and proof of the theorem, which is the engine for moving between probability measures.

- **Theorem Statement:** Let W_t be a standard Brownian motion under a probability measure P . Let θ_t be a process (the market price of risk) satisfying certain integrability conditions (Novikov's condition). Define a new process Z_t which is a P -martingale:

$$Z_t = \exp \left(- \int_0^t \theta_s dW_s - \frac{1}{2} \int_0^t \theta_s^2 ds \right)$$

We can define a new probability measure Q using Z_T as the Radon-Nikodym derivative: $\frac{dQ}{dP} = Z_T$. Girsanov's theorem states that the process $\tilde{W}_t$, defined as:

$$\tilde{W}_t = W_t + \int_0^t \theta_s ds$$

is a standard Brownian motion under the new measure Q .

- **Implication:** This powerful result allows us to transform a Brownian motion with a drift into a driftless Brownian motion by changing the underlying probability measure.

17 Stock Price under the Risk-Neutral Measure

Here, Girsanov's theorem is applied to the standard model for stock price dynamics.

- **Geometric Brownian Motion (GBM):** In the real world (under measure P), a stock's price S_t is assumed to follow GBM:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

Here, μ is the stock's expected rate of return (drift) and σ is its volatility.

- **Change to Risk-Neutral World:** We want to find a measure Q where the discounted stock price is a martingale, meaning its expected return is the risk-free rate r . We set the market price of risk $\theta = \frac{\mu-r}{\sigma}$. Using Girsanov's theorem, we define a new Q -Brownian motion $\tilde{W}_t = W_t + \theta t$. Substituting this into the GBM equation gives the stock price dynamics under the risk-neutral measure Q :

$$dS_t = r S_t dt + \sigma S_t d\tilde{W}_t$$

18 The Black-Scholes Formula

This lecture uses the risk-neutral framework to derive the famous Black-Scholes formula for a European call option.

- **Pricing Principle:** The price of a European call option, $C(S_t, t)$, is its discounted expected payoff under the risk-neutral measure Q .

$$C(S_t, t) = E_t^Q[e^{-r(T-t)} \max(S_T - K, 0)]$$

- **The Black-Scholes Formula:** Solving this expectation gives the explicit pricing formula:

$$C(S_t, t) = S_t N(d_1) - K e^{-r(T-t)} N(d_2)$$

where $N(\cdot)$ is the standard normal cumulative distribution function, and d_1, d_2 are given by:

$$d_1 = \frac{\ln(S_t/K) + (r + \frac{\sigma^2}{2})(T-t)}{\sigma\sqrt{T-t}}$$

$$d_2 = d_1 - \sigma\sqrt{T-t}$$

19 Final Lecture: Martingale Representation and Summary

The course concludes by stating a key theorem for hedging and summarizing the journey.

- **Martingale Representation Theorem:** This theorem states that any martingale (with respect to the filtration generated by a Brownian motion W_t) can be written as a constant plus a stochastic integral with respect to that same Brownian motion.

$$M_t = M_0 + \int_0^t \phi_s dW_s$$

This is the theoretical guarantee that in a Black-Scholes world (a complete market), any derivative can be perfectly hedged by continuously trading the underlying stock and the risk-free asset.

- **Course Summary:** The course provided a full journey from the basic principles of interest rates and portfolio theory in discrete time to the sophisticated machinery of stochastic calculus (Girsanov's theorem, martingales) required for pricing and hedging derivatives in continuous time, culminating in the Black-Scholes model.